



MOEMS®
Mathematical Olympiads for
Elementary and Middle Schools

PROBLEM OF THE WEEK 2025–2026

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6/29/26



When it is 7 AM in New York, it is noon in London. A plane leaves London at 12 PM (noon) London time and arrives in New York at 11 AM New York time the same day. A second plane leaves New York at 12 PM (noon) New York time for London.

What time in London is it when the second plane arrives?

Assume both planes fly for the same number of hours



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A purse contains 4 pennies, 2 nickes, 1 dime, and 1 quarter. Different values can be obtained by using one or more coins in the purse.

How many different values can be obtained?



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6/15/26



Two students are needed to work in the school store during the lunch hour every day, and four students volunteer for this work.

What is the greatest number of days that can be arranged in which no pair of the four students works together more than once?



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6/8/26



Twenty-four meters of fencing is used to fence a rectangular garden. Let M represent the number of square meters in the area of the garden.

What is the largest value that M could have?

Volume 1, Olympiad 49, #2, 5 Minutes



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6/1/26



Alice started a Math Club during the first week of school. As the only member, she decided to recruit two new members during the following week of school. Each new member, during the week following the week when he or she became a member, recruits two new members.

How many members will the club have at the end of five weeks?

Volume 1, Olympiad 35, #4, 5 Minutes



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5/25/26



A circular track is 1000 yards in circumference. Cyclists A, B, and C start at the same place and time, and race around the track at the following rates per minute: A at 700 yards, B at 800 yards, and C at 900 yards.

What is the least number of minutes it must take for all three to be together again?



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5/18/26

A		7	12
	4	9	
	5	16	
8	11		B

The figure below is a "magic square" with missing entries. When complete, the sum of the four entries in each column, each row, and each diagonal is the same.

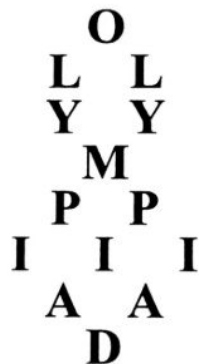
Find the value of A and the value of B.



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5/11/26



In this diagram, each “path” spells the word OLYMPIAD correctly.

How many different paths exist in the diagram?



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5/4/26



One light flashes every 2 minutes and
another light flashes every 7 minutes. If both
lights flash together at 1 PM,

**What is the first time after 3 PM that both
lights flash together?**

Volume 2, Set 6, Olympiad 3, 3D. 5 Minutes



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4/27/26



In one complete day, a man binds 100 books
and his helper binds one-fourth as many
books. If they take turns working complete
days,

**how many days would it take them to bind
500 books?**

Volume 2, Set 5, Olympiad 5, 5B. 4 Minutes



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4/20/26



Kim was elected class president. She received 3 votes for every 2 that Amy got. No one else ran. However, if 8 people who voted for Kim had voted for Amy instead, Kim would have received only one vote for every two that Amy would have gotten.

How many people voted?

Volume 2, Set 3, Olympiad 4, 4C. 5 Minutes



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4/13/26



This year, 1 out of every 55 adults started a new business. This rate represented an increase of 20% over last year's rate of 1 new business start-up out of every N adults.

Find N .



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4/6/26



Jess runs an outdoor stand at City Stadium. When it rains, Jess earns \$1500 selling umbrellas. But when it doesn't rain, she earns \$400 selling sunglasses. On any given day, the chance of rain is 40%.

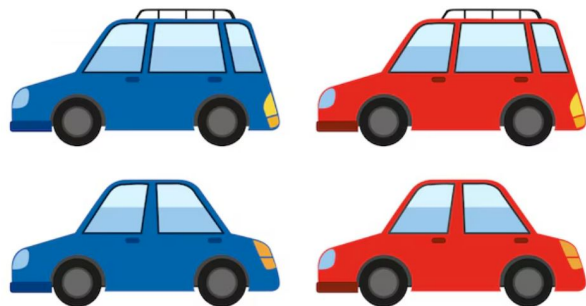
On the average, how much can Jess expect to earn daily?



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3/30/26



There are 2 red cars and 3 blue cars. The 5 cars contain a total of 12 people. No car has more than 4 people. Every car has at least 1 person. The only cars with the same number of people are the red cars.

How many people are in 1 red car?



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3/23/26

$$\begin{array}{r} \text{ONE} \\ + \text{FOUR} \\ \hline \text{FIVE} \end{array}$$

$$\begin{array}{r} \text{ONE} \\ \times \quad 4 \\ \hline \text{FOUR} \end{array}$$

In the following, different letters represent different digits.

What digit does the letter O represent?



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3/16/26



In a hotel, 40 people occupy exactly 26 rooms. Each room is occupied by either one or two people.

How many rooms are occupied by exactly one person?

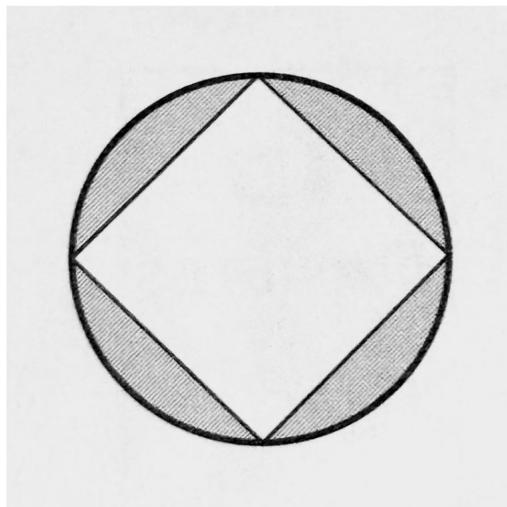
Volume 3, Set 15, Olympiad 1, 1B. 4 Minutes



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3/9/26



A square with an area of 18 square centimeters is inscribed in a circle as shown.

Using the approximation $\pi \approx 3.14$, find the area of the shaded region to the nearest square centimeter.



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3/2/26



In the repeating pattern below, what will be the 78th letter written?

ABBCCD ABBCCD...
(and so on)

Volume 3, Set 7, Olympiad 2, 2B. 4 Minutes



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2/26/26



Janine's number has three digits. One digit is a prime number. Another digit is a square number. The other digit is neither prime nor square. Her number is NOT divisible by 3.

What is the greatest possible value of Janine's number?

Volume 3, Set 4, Olympiad 1, 1D. 5 Minutes



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2/16/26



A pile of coins sits on a table. Sarah takes half of the coins plus 4 more. Then Nick takes 2. Then Joe takes 2 more than half of what is left. Finally Selena takes 5. Four coins remain on the table.

How many coins were on the table to start with?



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2/9/26



Three friends play a series of 8 games. For each game, the winner scores 8 points, the friend in second place scores 3 points, and the friend in last place scores 0 points. At the end of 8 games, Keri's score is 20 points.

In how many of the games did Keri finish last?

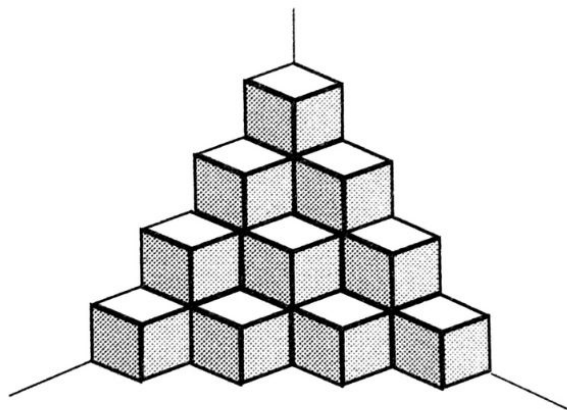
Volume 3, Set 7, Olympiad 1, 1C. 5 Minutes



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2/2/26



The tower shown is made of congruent cubes stacked on top of each other. Some of the cubes are not visible.

How many cubes in all are used to form the tower?

Volume 3, Set 4, Olympiad 4, 4C. 5 Minutes



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1/26/26



Mr. Alvarez gives each of his students 4 sheets of paper and 16 sheets are left over. But if two students were absent, each of the remaining students would receive 5 sheets, with only 3 sheets left over.

How many sheets of paper does Mr. Alvarez have?

Volume 3, Set 13, Olympiad 2, 2C. 6 Minutes



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1/19/26



Lou eats 1 jelly bean on September 1st, 3 on September 2nd, 5 on September 3rd, and so on, through the 30 days of the month. Each day he eats 2 more than the day before.

In all, how many jelly beans does Lou eat in September?

Volume 3, Set 10, Olympiad 5, 5C. 6 Minutes



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1/12/26



Ali and Ben have a total of 36 CDs. After Ali gives Ben 40% of her CDs, Ben has twice as many as Ali.

How many cds did Ali have originally?



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1/5/26



A bookseller has 15 different novels: 4 are in German, 5 are in Spanish, and 6 are in French. Emma buys two novels. they are written in two different languages.

In how many different ways can this be done? *Ignore the order in which she buys them*

Volume 3, Set 13, Olympiad 1, 1E, 7 Minutes



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12/22/25



A sports arena has a total capacity of 20,000 fans and ushers. One usher is required for every 30 fans.

What is the greatest number of fans that can be in attendance?



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12/15/25



Starting at the same time on opposite shores of a lake, two boats cross back and forth for 35 minutes without stopping. One boat needs 5 minutes to cross the lake. The other boat needs 7 minutes.

What is the number of times during the 35 minutes that the faster boat passes the slower boat going in the same or opposite direction?

Volume 3, Set 11, Olympiad 5, 5D, 7 Minutes



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12/08/25



A package weighs P pounds, P being a whole number. To ship this package by express costs \$1.65 for the first five pounds and 12¢ for each additional pound. The total shipping cost was \$3.45.

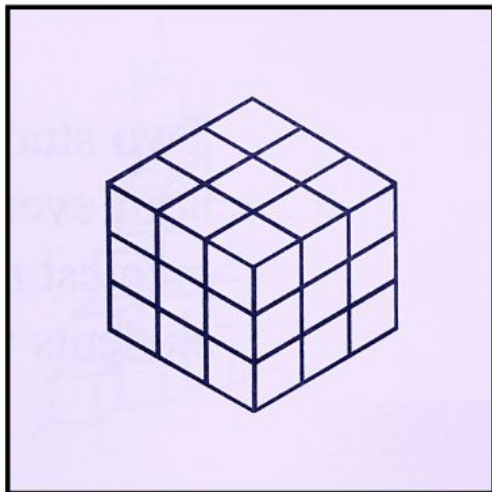
How many pounds did the package weigh?



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12/01/25



Below is a 3 by 3 by 3 cube. Not all of the cubes are visible. Suppose the entire outside of the cube is painted red including the bottom.

How many different 2 by 2 by 2 cubes with exactly three red faces can be found in the shown cube?



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11/24/25



The cold water faucet of a bathtub can fill the tub in 15 minutes. The drain, when opened, can empty the full tub in 20 minutes. Suppose the tub is empty and the faucet and drain are both opened at the same time.

How long will it take to fill the tub?



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11/17/25



A quart of milk can feed either 6 cats or 10 kittens. Suppose there are 3 quarts of milk and 15 cats.

After all the cats are fed, how many kittens can be fed with the leftover milk?

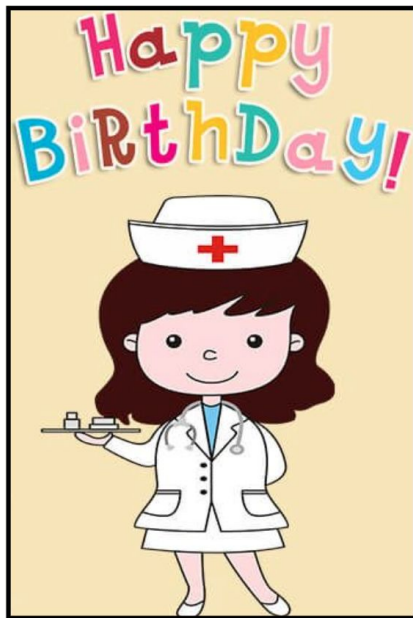
Volume 2 Set 15, Olympiad 5, 5A. 4 Minutes



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11/10/25



Dr. Bolton was born in an interesting year.
The tens digit was twice the thousands digit,
the ones digit was three times the tens digit,
and the hundreds digit was equal to the sum
of the other three digits.

In what year was she born?

Volume 2 Set 6, Olympiad 3, 3A. 3 Minutes



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11/3/25



At Hudson High School, the first class starts at 8:26 AM and the fourth class ends at 11:26 AM. There are 4 minutes between classes and each class is the same length.

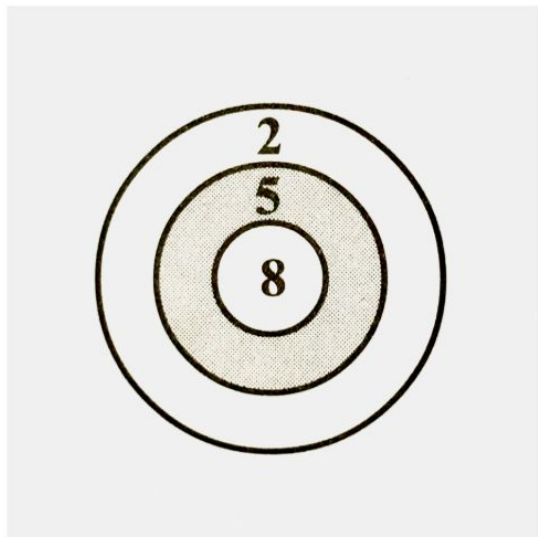
How many minutes are there in one class period?



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10/27/25



Each of three darts lands in a numbered region of the dart board, scoring the number of points shown.

How many different sums are possible for the three darts?

Volume 2, Set 11, Olympiad 1, C. 5 Minutes



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10/21/25



A gumball machine contains red, green, yellow, and purple gumballs. You cannot control which color you get. Anna wants three gumballs of any one color.

At 5¢ each, what is the minimum number of cents that guarantees in advance three gumballs of the same color?

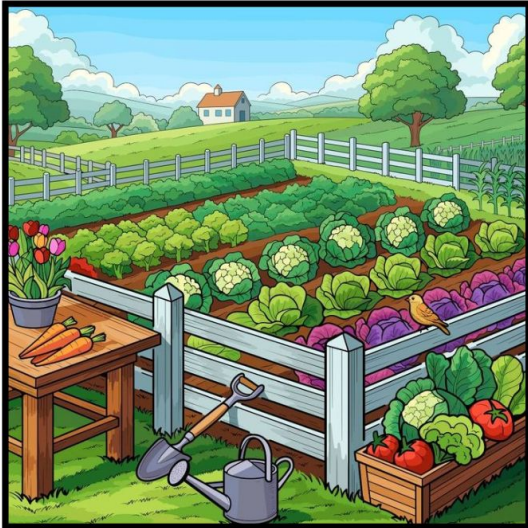
Volume 2, Set 7, Olympiad 4, 46. 5 Minutes



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10/13/25



A fenced rectangular garden is 10 m wide and 20 m long. When one side is moved outward and two other sides are increased in length, the area increases by 40 sq m.

What is the fewest number of meters of additional fencing needed to form the larger rectangular garden?



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10/6/25



Jessie has \$5.10 worth of stamps. She has equal numbers of 50-cent, 20-cent, 10-cent, and 5-cent stamps. She has no other stamps.

How many 50-cent stamps does she have?



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9/29/25



A single story house is to be built on a rectangular lot 70 feet wide by 100 feet deep. The shorter side of the lot is along the street. The house must be set back 30 feet from the street. It also must be 20 feet from the back lot line and 10 feet from each side lot line.

What is the greatest area that the house can have, in sq ft?

Volume 2, Set 5, Olympiad 5, 5D. 6 Minutes



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9/22/25



Amanda, Beth, and Sarah run three races. In each race, one of them earns 5 points, one of them earns 3 points, and one of them earns 1 point. After the three races Beth has the highest point total.

What is the smallest total score that Beth can have?

Volume 2, Set 10, Olympiad 2, 2C. 6 Minutes